Midterm Project: Classification in Statistical Learning

Course: Introduction to Statistical Learning

Due Date: 10/25/2026

Format: Individual or Group (up to 2 students)

1. Objective

This project evaluates your ability to apply classification methods to real-world data. You will compare methods, assess model performance with cross-validation, and interpret the results. By the end, you should be able to:

- Fit and compare several classification methods.
- Use CV to estimate prediction error.
- Communicate findings clearly in both technical and plain language.

2. Examples of Datasets - UCI Machine Learning Repository

- Breast Cancer Wisconsin Diagnostic (UCI): Classify malignant vs. benign tumors.
- Finance: Credit Card Default (UCI): Predict default vs. non-default.
- Chemical analysis: Wine Quality (UCI): Binary classification (quality ≥ 6 vs. < 6).
- Choose your own research topic and choose features that you are interested in.
- Kaggle: https://www.kaggle.com/datasets?tags=13302-Classification

3. Project Tasks

Part A. Data Preprocessing (10 Points)

- Discuss preprocessing: missing values, scaling, categorical encoding.
- Outlier detection and handling

Part B. Data Exploration (20 points)

- Describe the dataset (number of observations, predictors, target distribution).
- Perform basic EDA: plots, summary statistics, correlations.

Part C. Classification Models (50 points)

- Choose at least two classification methods but not limited to the ones we covered so far.
- Compare predictive performance using CV.

- Report accuracy, precision, recall, and ROC/AUC.
- Discuss similarities/differences among methods.

Part D. Synthesis & Discussion (10 points)

- Which classification method performed best overall? Why?
- (Optional) How did resampling influence your understanding of error rates?
- Discuss practical takeaways: interpretability vs. prediction accuracy.

Part E. Deliverables (10 points)

- Report (up to 5 pages, PDF): Abstract, introduction, methods, results, discussion/conclusion. Include key figures/tables (coefficients, CV results, ROC curves, etc.).
- Code (Jupyter Notebook or R Markdown/Qarto): Clean, reproducible, well-documented. If it exceeds page limit, please place it to Appendix.